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Macroeconomic Uncertainty and Economic Value Creation: An Agent-Based Simulation of Firm Behavior Under Financial Shocks --Manuscript Draft--

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Macroeconomic Uncertainty and Economic Value Creation: An Agent-Based Simulation of Firm Behavior Under Financial Shocks

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Abstract

Increased frequency of financial crises highlights the need to learn more about the ways companies learn and create value in the face of macroeconomic uncertainty. In this work, an Agent-Based Macroeconomic Model (ABM) is developed to look at the behavior of firms during a financial shock with liquidity freezes, credit rationing, and policy instability. As opposed to the conventional equilibrium-based models, the model comprises heterogeneous agents; firms, banks, households, central bank and government; whose behavior is determined by stochastic rules in response to uncertainty shocks and nonlinear feedbacks between the real and financial sectors. The model emphasizes the creation of economic values as a result of adaptive actions taken by firms in the area of investment, production and risk management. The model explains a big part of macroeconomic and financial stylized facts through endogenous money creation mechanisms, precautionary borrowing and regulatory restrictions. They include the pro-cyclicality of investment and credit, counter-cyclicality of bankruptcies, right-skewness of the duration of banking crises, fat-tailed fiscal costs and Phillips and Okun relationships. The results of simulations emphasize how uncertainty shocks amplify volatility, extend recession, and heighten systemic vulnerability, but that such policy interventions as fiscal transfers and central bank liquidity injections can reduce but not eliminate instability. This contribution introduces a generic, calibrated ABM method to explore firm level adaptation in cases of uncertainty and provides new insights into how macroeconomic shocks contribute to resilience, systemic risk and long-run value creation.

Keywords: Macroeconomic uncertainty; Agent based modeling; Firm behavior; Financial shocks; Economic value creation; credit cycles; financial stability

1. Introduction

Financial system is a complex and dynamic system, volatile in nature and vulnerable to fragility and disruption. During the past few decades, the financial system has witnessed numerous events such as the 2007/2008 financial crisis as an antecedent to reorganizing the financial system resilience and adaptability to endogenous and exogenous shocks (Melina et al., 2023). Such shocks which are often unpredictable in nature unveil fundamental vulnerabilities in the system most prominently in businesses and their ability to survive periods of high uncertainty (Ma et al., 2024). It is crucial to understand how the firms act in such uncertain situations to assess economic stability and value creation as a reaction to those risks. The focus of this article is macroeconomic uncertainty, the economic conditions that cannot be predicted in advance, and how companies can be innovative and create value during fiscal instability (Haynes & Alemna, 2022). The main point is to model the adaptive behavior of firms to the financial shocks, and concentrate on the activities of decision-making by firms regarding investment, production, and financial control in the uncertain condition (Oztemel & Isik, 2025). Unlike the conventional models, which rely on deterministic assumptions, the research incorporates stochastic components, which resemble the nonlinear dynamics that characterize actual financial crisis such as liquidity freezing, market failure, and instant credit rationing. Even though the monetary system is essential to the achievement of financial stability, its influence on the behavior of the firm in circumstances of uncertainty is not a major concern of the existing models (Akhatoeva et al., 2022). The economic activity is founded on the financial system, its credit-creating and money-circulating processes, and institutional setting (Giannakopoulos et al., 2024). The relationship between macroeconomic uncertainty and the decision-making of firms in such systems, however has been poorly investigated, especially when subjected to financial shocks. The current debate on financial reform and stability has been driven by the focus on systemic risk and the need to have good mechanisms in place so as to buffer against shocks. Scarcely anything is said, however, about the more general question: How do companies create value in a financially uncertain world? The current paper aims to address that gap by analyzing firm behavior in a situation of financial uncertainty and its impact on the generation of economic value (Ionescu et al., 2023). It integrates the results of agent-based modeling (ABM) to simulate the decision-making at the firm level and the aggregate effect on the economy overall during financial crises. In that spirit, we use an Agent-Based Macroeconomic Model (ABM) that incorporates interdependencies between heterogeneous (households, banks, firms) and adaptive responses to economic shocks (McGarrahy et al., 2022). ABMs are well-placed to achieve this since they allow inclusion of nonlinear feedback mechanisms, which are a key component to explain cascade dynamics of financial instability (Ionescu et al., 2025). They are ill-modeled using traditional models like Dynamic Stochastic General Equilibrium (DSGE) because it typically assumes rationality and equilibrium conditions, which cannot be true in periods of crisis. A special feature of this model is that it allows modeling macroeconomic uncertainty and financial shocks in a dynamic and interactive framework (Keenan et al., 2021). This study offers a comprehensive view of the complexity of economic systems under stress with the inclusion of the reaction of firms in financial distress and the efforts to adapt and rationalize their efforts at creating value. The model extends the traditional models by the inclusion of stochastic variables which reveal the uncertainty dynamics of a financial market and allow more realistic explanations of the survival of firms, their investment behaviour, and market performance. The paper makes contributions to the literature in two basic ways. First, it is a methodological contribution as an agent-based framework is developed that can be further extended and calibrated under a variety of macroeconomic parameters. This framework is flexible enough to accommodate changes in economic environment and can be used to analyse different types of financial shocks and their impact on business and economy in general. Second, it has an economic impact through introducing new knowledge to macroeconomic uncertainty, firm behaviors and value creation in the economy. The paper illustrates how companies can cope with uncertainty and keep on generating value even when faced with extreme financial shocks by modeling different crisis scenarios.

Contribution to Research

- Possible Research Topic: Model Company Behavior in Macroeconomic Uncertainty: To develop an agent-based simulation model, in which firms modify their strategies (i.e., investment, production, and financial management) in the presence of macroeconomic uncertainty and financial shocks.
- Model Financial Shocks and Their Effects: To explore how various kinds of financial crises (e.g., liquidity crises, interest rate shocks, market shocks) can impact the economic value creation of firms with a focus on the impact of uncertainty on firm survival, profitability, and growth.

- **Study Value Creation in the Presence of Uncertainty:** To study how firms can sustain or increase economic value creation in presence of uncertainty generated by financial shocks by considering such variables as risk management, innovation and adaptive behavior in crisis scenarios.
- **Embed Nonlinear Feedbacks and Systemic Risk:** Producing a model using an agent-based model to simulate the nonlinear feedback between the financial markets and the real economy that allows one to understand how firm interactions and systemic risk spread throughout the economy when the financial markets are experiencing a financial stress.
- **Provide Policy Recommendations to Economic Stability:** To provide valuable information and policy recommendations to promote better economic stability, resilience, and growth, in particular how monetary policy, regulatory policy and financial reforms can better equip firms to flourish in unpredictable times.

The paper is organized as follows: Section II provides an introduction to the materials and methods, which describe the agent-based model and the elements of this model. Part III deals with the economic conditions and model requirements, especially, how firms behave during confusion and value-generation processes during financial shocks. Section IV is the calibration and validation of the model to assure that the model is a description of the actual economic processes. Part V ends with the introduction of the results of the simulation and the conclusions regarding both theoretical and practical implications of the work in the sphere of macroeconomic policy.

2. Literature Review

The interplay between the creation of economic values, firm behaviour and macroeconomic uncertainty has increasingly attracted the attention of economics, in recent years in particular given the global financial crises and growing concern about the interdependence of financial and real economic activity. Key contributions to this area described in the current review include the theory of firm behavior in the face of uncertainty, the role of financial shock, and agent-based modelling (ABM) as a method to predict the dynamics.

2.1 Firm behaviour and Macroeconomic uncertainty.

The concept of macroeconomic uncertainty has been researched in several studies, especially in the context of financial crisis, economic market dynamics and economic policy changes. The key point that (Zhou & Shen, 2024) highlights is the fact that uncertainty contributes to the investment behaviour of firms. According to his findings, more uncertainty can lead to less investment because companies will not be willing to make long term commitments when the economic situation remains uncertain. Another factor that (Dorrah & McCabe, 2024) emphasize is that uncertainty about the future demand, together with the lack of finance, has a significant impact on the firm-level decision-making, particularly when it comes to the policies of capital expenditure and human resources recruiting. Several studies have been conducted to understand the responsiveness of the companies to uncertainty as adaptive responses. Positively, (Mazzetto, 2024) assume that organizations delay investment and reduce the scale of their production potentials in response to the increase in uncertainty, leading to a decrease in the level of growth and making them very vulnerable in the event of a financial crisis. (Peykani et al., 2025) consider macroeconomic outcomes of firm-level uncertainty and find that persistent uncertainty on a company level may result in the overall economic stagnation.

2.2 Economic Value Creation and Financial Shocks.

Financial shocks (credit crunches, liquidity crises, or sudden changes of market sentiment) can have profound effects on the ability of firms to create value. (Demopoulos, 2025) depict the effects of financial shocks that change firm behavior through mechanisms such as credit rationing and liquidity freezing has a direct effect on firms, reducing access to financing, and, therefore, reducing its ability to invest and produce. (Martins et al., 2023) discuss the economic processes related to the meaning of financial crises and the ways in which the shocks in the financial system, e.g., fluctuations in the availability of credit, can propagate in the economy, influencing the survival and profitability of firms. On the other hand, (Gambino et al., 2024) examine how financial crises may become the trigger that prompts firms to engage in strategic change, which would result in new value creation strategies. As an illustration, in recessions, firms shift their focus on to more resilient forms of businesses, resource efficiency or inventing new kinds of products and services to conform to the unpredictability. Such flexibility can offer novel sources of economic value in spite of severe financial shocks.

2.3 Agent-Based Models (ABM) in Macroeconomic Simulations

The idea of agent-based modeling (ABM) has emerged as a useful approach to the uncovering of the complex dynamics of agents within economic systems, in particular in the case of macroeconomic uncertainty and financial shocks. Representative-agent models including Dynamic Stochastic General Equilibrium (DSGE) focus on representative agents and equilibrium, which are lacking in the light of non-linear dynamics or systemic risks associated with financial instability. ABMs, in their turn, are well-adapted to nonlinear interaction of firms, households and banks, as well as capturing heterogeneity. The first pioneering research established by (Sun et al., 2025) illustrated the ability of ABMs to model economic agents of limited rationality and adaptive behavior in ways that allow the emergence of such complex economic events like a financial crash. More recently ABMs have been applied by Dosi et al. (2010) to model the business cycle showing how non-linear feedback effects, including market crashes or liquidity spirals, are caused by firm-level heterogeneity and financial shocks. A number of studies have adopted ABMs specifically to analyse the behaviour of firms during financial shocks and macroeconomic instability. (Guyen et al., 2025) developed an ABM that models the influence of financial crises on investment decisions of firms and discovered that the investment decision of firms largely depends on the relationship with banks and access to credit. Similarly, (Wang & He, 2024) employed ABMs to determine the propagation of financial shocks in the markets across the entire real economy to determine their effects on the performance of firms and macro-economic stability. Moreover, (Wang & He, 2024) further used ABMs to study the impact of macroeconomic uncertainty on the decisions of firms in terms of investment and pricing. This model that incorporated adaptive expectations

indicated that firms actions can have diverse macroeconomic effects depending on the degree of uncertainty and occurrence of financial shocks.

2.4 ABMs and Systemic Risk and Feedback Mechanisms.

The general term in macroeconomics and financial theory used to define risks that are faced throughout the financial system and may destabilize the economy is systemic risk. According to the Bank of England (2014) and Borio (2009), systematically modeling the risk is essential in the interpretation of the broader economic implications of financial instability. ABMs are especially good at modeling systemic risk, because they can model nonlinear feedback loops and agent interactions critical to learning how crises emerge. The article by (Rezaei et al., 2025) shows how heterogeneously interacting agents (e.g., firms and banks) can generate financial contagion in the sense that the collapse of one agent (e.g., a firm or a bank) can trigger the destabilization of the economy on a significantly bigger scale. This self-referential feedback process is the core of ABMs and makes them helpful particularly in investigating financial shocks and how the same can propagate to the rest of the economy. Moreover, (Simmons et al., 2021) have defined that the feedback mechanisms that can exist between the firm decisions, market moods and financial restrictions have the capacity of amplifying the market shocks which exacerbate the economic downturns. ABMs are especially better placed to capture such dynamics in terms of modeling the interaction between agents whose expectations are adaptive and have different strategies depending on market conditions and access to information.

2.5 Why Agent-Based Models are required in the study of firm behavior during a financial shock.

Although the issue of macroeconomic uncertainty and financial shocks as determinants of the behavior of firms is getting more and more attention, the literature has not covered adequately those models that can give a comprehensive explanation of the actual realities of decision-making at the firm level when in crisis situations (Serkov & Krasnykh, 2024). Most existing models such as the DSGE focus on equilibrium and representative and rational actors and are not based on the non-stationary, stochastic aspects of economic systems in a crisis. The use of agent-based models (ABMs) offers a solution to this shortcoming in that they model heterogeneous agents interacting according to simple adaptive rules, giving rise to emergent macroeconomic dynamics. As this review illustrates, ABMs have been applied with success to model financial crises, systemic risk, and firm behavior under uncertainty but their potential has not been fully exploited in the study of macroeconomic uncertainty and economic value creation in the face of financial shocks.

3. Materials and Methods

3.1 The Modelling Language for Linked Lives (ML3)

We use the Modeling Language for Linked Lives (ML3), an adaptable framework initially developed for agent-based demographic simulation, to implement the model for simulating firm behavior under macroeconomic uncertainty and financial shocks. Although ML3 was originally designed to model interactions among people, its flexibility and structure make it suitable to easily transform into modeling economic processes, especially in the modeling of firm behavior when facing uncertainty. Firms' economic value creation process, as well as the way they react to financial shocks, is mapped onto the concepts of ML3, making it possible to simulate intricate, dynamic economic systems. Central to ML3 is the stochastic rule concept, describing the behavior of agents within a probabilistic setting. Within our adapted application, agents are firms, households, and banks that have their respective behaviors and interactions. The agents respond to variations in the macroeconomic scenario, which could be surprise financial shocks, liquidity crises, or extremely volatile market conditions.

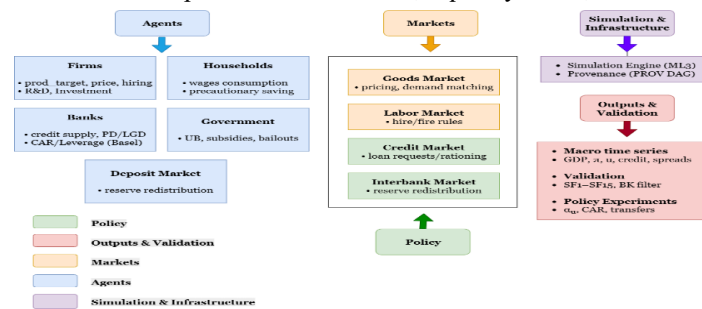


Fig. 1. Agent-Based Structure of the Economic Simulation

Every rule in ML3 has three components:

- 1) Guard: The guard condition identifies the category of agent to which the rule is to be applied. For example, a rule for making investment decisions would be applicable to companies when levels of uncertainty are higher than a specified value.
- 2) Rate: The rate signifies how often the rule will be executed. It can be represented by an exponential distribution to capture the random nature of financial shocks and uncertainty. For instance, the triggering of a financial shock (e.g., an unexpected liquidity crisis) could be initiated at a randomly established rate, capturing the random nature of economic disruptions.
- 3) Impact: The impact specifies the action of an agent when a rule is being implemented. For companies, this may be adapting their investment pattern, hiring strategy, or production policy in accordance with the prevailing economic environment. For example, during an economic contraction or liquidity shock, companies may cut investment, cut down on production, or even postpone hiring. The stochasticity of such decisions is critical to the capture of firm-level uncertainty and adaptive behavior within an unstable environment.

In the simulation, every agent's action is performed according to the minimum waiting time from the stochastic rules, moving the simulation forward by the quantity of time as indicated by these rules. This provides a dynamic, non-linear path of firm behaviors with changing economic conditions where uncertainty and shocks are in focus. The ML3 infrastructure separates

the model logic (Fig 1) (the rules that define how the agents act), and the simulation logic (how to execute these rules), making the model easy to modify and extend. The modularity plays a vital role in exploring a wide range of situations of macroeconomic uncertainty and financial distress in which they will act variously as the situation changes.

3.2 Provenance

To record the simulation model and make its economic basis transparent, we combine a provenance model with the textual model description. **The provenance model tracks the source of the elements that were used in the simulation, displaying their basis in economic theories and indicating the process followed in developing the model.** This approach enhances the transparency of the model's assumptions and limitations so that the generated results are based on well-established economic principles. The provenance model is constructed from the PROV standard and describes the development process as a directed acyclic graph (DAG). Entities (represented as circles) in this graph denote the different inputs, such as data, economic theories, and previous models, and the resulting products of the simulation, such as the components of the model and the results of the simulation. Activities (represented as squares) denote the processes that connect these entities, such as the modeling process, rule combination, and simulation experiments. The provenance graph clearly delineates how economic theories, including those concerning financial shocks and macroeconomic uncertainty, are connected with agents' behavior in the model. For example, the firms' decision-making process is guided by uncertainty theories, while the banking sector works according to liquidity risk models. This makes the agents' behavior in the simulation non-arbitrary but based on existing economic theories. By the inclusion of the provenance model, we render the process of simulation and the basis on which it is done more transparent. This makes it easier to communicate how macroeconomic uncertainty affects firm conduct and facilitates checking the robustness and generalizability of the simulation outcomes across various economic scenario.

3.3 Model Outline

As outlined in the introduction, this paper introduces a methodological extension of macroeconomic agent-based models (MABMs) based on macroeconomic uncertainty and how it affects economic value creation by companies in the context of financial shocks. The model seeks to investigate firm conduct under financial uncertainty and how companies respond to macroeconomic movements and related financial instability. The specifications for this model are given in the subsequent paragraphs.

3.3.1 Model Needs

In order to quantify and compare macroeconomic stability and financial stability at the general level, the interrelationship between the real and financial sectors needs to be captured. Such interaction determines credit demand, investment activities, and value creation in the economy during times of financial disturbances. This model, hence, has feedback connections between these sectors that reflect adaptive firm behaviour due to macroeconomic uncertainty.

The model requires the following:

1. Business Cycle (Requirement 1):

- The model consists of five key elements, the main component of which is the description of **the dynamics of the real sector, relationships between households and firms.** These interactions create influence on the credit demand and supply and in their turn, they influence the overall performance of the economy. The model uses the business cycle to reproduce the volatility of firm-level investment behaviour and households' consumption patterns both of which are caused by macroeconomic shocks. The model must consider macroeconomic performance variables which are GDP, employment rates, inflation and the survival rates of firms during crises.
- Adaptation to Uncertainty: Macroeconomic uncertainty guides investment and labor decisions of firms, such as fickle market demand or change in the finance system. This adaptation process is where the simulation of value creation in uncertain conditions is simulated.

2. Financial Cycle (Requirement 2):

- The credit cycle is modelled by the financial cycle and its interaction with the real economy. **Financial shocks, e.g., liquidity crises, are at the core of influencing firms' access to investment funds and credit.** Banks' credit supply decisions (e.g., firm loans) are influenced by the shocks, which can result in overshooting credit cycles and hence amplifying systemic risk.
- The model focuses on how financial uncertainty influences credit behavior, potentially leading to unsustainable credit growth or credit rationing during crises. A key consideration is whether credit is productive (increasing firm value) or unproductive (rolling over debt, leading to higher default risks).
- Systemic Risk: While companies face financial uncertainty, the model will explore how systemic risk arises through the interactions of firms' credit demand, credit creation, and the overall economy. This is essential in learning how shocks affect the financial system, driving aggregate economic value.

3. Firm Behavior Under Uncertainty (Requirement 3):

- In this framework, companies are agents who have to make strategic choices on investment, production, and employment faced with macroeconomic uncertainty. The credit provision, financial stability and the degree of economic uncertainty at the market will dictate their actions.
- 4. Risk Management: Businesses will be forced to adapt to the environment of financial shocks, re-setting the intensity of production and investments in the light of uncertainty in demand and the credit availability. The model is emphasizing the way companies have dealt with financial risks like liquidity risks, credit default risks, and changes in interest rates as core of their survival during the financial shocks as shown in the Fig 2.

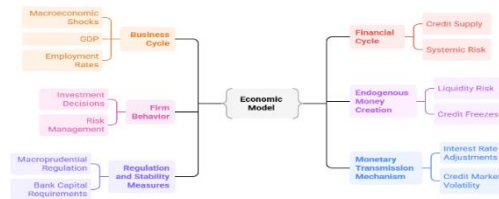


Fig. 2. Conceptual Framework of the Economic Model

4. Endogenous Creation of Money and Financial Shocks (Requirement 4):

- The model contains the exogenous money creation within the financial systems. The credit generated by banks is used by companies to perform production and investment. Financial shocks can cut off the process, though, because fluctuations in the supply of credit directly affect the ability of firms to fund their operations.
- During financial shocks, liquidity risk is the main one. This model will be a way of reflecting the reaction of the banks and firms to the shortage of liquidity, which might result in freezing credits and the collapse of the real economy. It is this cycle of investment patterns, corporate demand of credit and money creation that is fundamental to understanding how value can be destroyed or created in the conditions of financial uncertainty.

5. Measures of Regulation and Stability (Requirement 5):

- Macroeconomic policy and regulation are essential in the occurrence of systemic risk, which occurs due to financial instability. The model has a macroprudential regulation system, modelled after Basel III and other financial regulation systems to encompass the effect of regulatory policy innovations on the accessibility of credit by firms and overall economic stability.
- Financial Regulation in Crisis: During financial shock period, the model will look at the impact of capital requirements of banks, liquidity ratios and other regulations to supply of credit and stability of firms. The model can be used to test the implication of financial regulation on economic value creation in crisis situations by modelling different regulatory environment.

6. Monetary Transmission Mechanism (Requirement 6):

- The monetary transmission mechanism explains the impacts of monetary policy adjustments (e.g. interest rate adjustments) on investment intentions by firms and the supply of credit. This is significant in establishing ways in which the impact of financial shocks can be reduced or enhanced by the monetary policy.
- Interest Rates and Firm Behavior: The model is based on the influence of interest rates changes on access to finance by firms which in turn affects investment decisions by firms and recruitment decisions. The central bank monetary policy in situations with high uncertainty or financial shocks can stabilize or rather enhance volatility on credit markets.

3.4 Model Overview

The designed framework puts companies at the core of the model, as they are the main creators of economic value and most exposed to macroeconomic uncertainty. Companies decide on production, investment, employment, and financing, and heterogeneity (size, productivity, and risk aversion) among them determines how the shocks transmit in the system. Surrounding institutions—families, banks, the central bank, and the state—act on firms in a manner that transmits and magnifies uncertainty, generating feedback loops between the real and financial economy. These interactions play out through five major markets: goods, labor, credit, deposits, and interbank. Demand shocks have an impact on firm revenues and hiring decisions in the goods and labor markets, while banking behavior when uncertain determines access to credit in the credit and deposit markets. The interbank market is a stabilizer or amplifier of shocks based on liquidity conditions. The markets as a group connect firm-level uncertainty to aggregate macroeconomic outcomes as illustrated in the Fig 3.

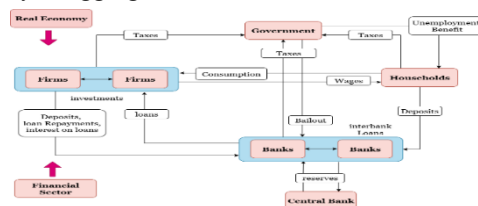


Fig. 3. Flow of Transactions Between Real Economy and Financial Sector

as currently written, the economic model is not sufficiently grounded in microeconomic principles, several core objects are undefined or inconsistent, and the "validation" relies on stylized-fact matching that is not informative given the degree of discretion in rule design and calibration.

Characteristic of the framework is clear modelling of uncertainty shocks, including demand contraction, liquidity freezes and unpredictability with regard to policy. Such shocks disrupt usual decision-making and make the firms dynamic. Other companies react to this by reducing their expenditure or reducing investment, and some even collapse completely, causing bankruptcy. The system thus averts attention to the duality of value creation or value destruction, which depends on the reaction of firms and institutions to shocks as well as the policies enforced to restore stability.

Key Elements

- Agents: Firms (focus), households, banking, central bank, and government interacting within a macro-financial system.
- Markets: Interbank markets, good, labor, credit, deposit and interbank markets between the real and financial sectors.
- Firm Behavior: Adaptive behavior under uncertainty (investment, pricing, hiring or default).
- Uncertainty Shocks: Decreased demand, liquidity paralysis, and uncertain policy-making that re-establishes firm results.

5. Value Creation vs. Destruction: Aggregate performance is determined by the ability of firms to absorb shocks and the performance of institutions in absorbing the risk of a systemic crisis.

3.4.1 Goods Market

The real economy comprises two submodels, the goods market and the investment market, that jointly model the business cycle in conditions of macroeconomic uncertainty (requirement 1 in Section 3.1). The four mechanisms by which the goods market is structured are (1) the supply behavior of the firms, (2) the dynamic of the labor market, (3) household demand and (4) the firm bankruptcies. It is these mechanisms that collectively influence the creation or destruction of economic value in times of financial shock and uncertainty among firms.

3.4.2 Supply mechanism of firms

Companies adjust their production and pricing patterns to demand impulses that are susceptible to shocks. Household consumption (Section 3.3.3) and firm investment (Section 3.4) create demand. The model of production capacity is:

Production capacity

$$Y_i = \Lambda_i \cdot \alpha \quad (1)$$

Y_i = production capacity of firm i , Λ_i = the level of labor hired, and α = fixed labor productivity.

The companies will determine a production level Y_i and a price p_i , which will be renewed based on the demand situations and uncertainty shocks:

Rules of production target and price update: —> where do we have this rules?
This is not derived from profit maximization under any explicit demand system, technology, or price-setting environment.

Production target and price update rules

$$Y_i^T = Y_i \cdot [1 + \gamma_y \cdot \xi], \text{ if } Y_i < D_i \text{ and } u > u^{nat} \quad (2)$$

$$p_i' = p_i \cdot [1 + \gamma_p \cdot \xi], \text{ if } Y_i < D_i \text{ and } u < u^{nat} \quad (3)$$

$$Y_i^T = Y_i \cdot [1 - \gamma_y \cdot \xi], \text{ if } Y_i > D_i \text{ and } p_i \leq \bar{p} \quad (4)$$

$$p_i' = p_i \cdot [1 - \gamma_p \cdot \xi], \text{ if } Y_i > D_i \text{ and } p_i > \bar{p} \quad (5)$$

Here:

- D_i = demand for firm i 's goods,
- u = aggregate unemployment rate,
- u^{nat} = NAIRU threshold, —> was not explained
- $\bar{p}$ = average market price,
- γ_y, γ_p = sensitivity parameters,
- $U[0,1]$ = stochastic uncertainty shock. —> can't figure it out in the model

In such a way, boundedly rational firm strategies integrate both shock volatility and bounded rationality, which influence the outcomes of value creating under uncertainty.

Firm rule — uncertainty-aware strategy & value-creation safeguard

Firm

| true

@ ego. strategy change rate () -> // Update production target & price with uncertainty and value-creation checks

if (gap > 0) then

ego. prod target: = ego. output * (1 + ego. gamma y * (1 + xi))

if (inv ok = false) then

ego. price: = ego. price * (1 + ego. Gamma p * xi) —> why it is not 1+xi ?

end

else

ego. prod target: = ego. output * (1 - ego. gamma y * (1 + xi))

if (p gap < 0) then

ego. price: ego. price * (1 - ego. Gamma p * (1 + xi))

end

end // Resilience cutbacks when risk is high or value creation turns negative

if (risk > ego. risk threshold) or (vc < 0) then

ego. Prod target: max (0, ego. prod target * (1 - ego. caution))

ego. Hiring freeze: true

end

were

xi: Shock Manager. Demand uncertainty () // ~ U[0,1]

gap: ego. demand - ego. output // $D_i - Y_i$

p gap: Mercury. average Price () - ego. price // $\bar{p} - p_i$

inv ok: (ego. inventory >= ego. inventory target)

risk: Risk. score (ego. debt ratio, ego. Interest burden, xi)

vc: Accounting. Value creation (ego. revenue, ego. costs, ego. wacc) it is not formally defined in the economic model section and it is unclear how they map into microeconomic primitives

The initial ML3 rule captures how firms vary their production and pricing plans in the face of macroeconomic uncertainty. Firms react to gaps in demand through the raise or lower of their production targets, with price revision being a function of the relationship between their own price and average market price. These changes are supplemented by a stochastic

uncertainty factor (ξ_i) which shows volatility as an indicator of demand shocks. It is noteworthy that the regulation introduces a value-creation insurance: at the point at which the risk rating of a company is too high or the net value creation turns negative, the company will reduce production and cease filling vacancies. This describes how firms attempt to keep afloat and evade losses during financial shocks and essentially bridges the gap between micro-level choice and aggregate patterns of resilience and fragility. → This means the firms face no loss and they know exactly how demand shocks work?

3.4.3 Labor market

In order to satisfy the production targets, firms' layoff or recruit workers according to the difference between labor demand L_i^T and actual Λ_i . The adjustment is:

Labor market adjustment

$$\text{hire rate} = z_h \cdot \frac{L_i^T - \Lambda_i}{\Lambda_i}, \text{ if } L_i^T > \Lambda_i \quad (6)$$

$$\text{fire rate} = z_f \cdot \frac{\Lambda_i - L_i^T}{L_i^T}, \text{ if } L_i^T < \Lambda_i \quad (7)$$

where z_h and z_f are sensitivity parameters. In financial shocks, precautionary firings can take place, increasing uncertainty and lowering aggregate demand.

3.4.4 Demand mechanism of households

Household consumption depends on disposable income (wages and benefits). Households spend income on goods but cut spending as uncertainty rises, exhibiting precautionary saving behavior. Consumption is simulated through a stochastic matching mechanism, with households choosing goods from firms depending on stock and price:

Household consumption

Household block is extremely underdeveloped → no intertemporal choice, no explicit precautionary saving mechanism as mentioned, no budget evolution beyond the period

$$C_h = \min\{W_h + B_h, \sum_i p_i \cdot q_i\} \quad (8)$$

where C_h = household consumption, W_h = wages, B_h = benefits, p_i = price, q_i = quantity.

The matching employs weighted random selection in proportion to $1/p_i$ to reflect bounded rationality with incomplete price knowledge. → We don't care about their survival when the basic consumption didn't meet? And how can we compute individual demand when $W_h + B_h$ decide consumption?

3.4.5 Firms' bankruptcy

A firm becomes bankrupt if its debt ratio is over an insolvency level:

Bankruptcy condition

$$\frac{\text{Debt}_i}{\text{Equity}_i} > \theta \quad (9)$$

with θ = default barrier (e.g., 200%). After becoming bankrupt, companies close credit lines, fire employees, and leave the market. This diminishes economic value both directly (by means of missed production) and indirectly (by means of banks' equity stress). During times of financial shocks (liquidity crunches, increasing borrowing prices), bankruptcies spike strongly, showing how uncertainty results in systemic value destruction.

Bank Credit Decisions during Liquidity Freezes

Bank

| firm.loan_request > 0

@ ego.credit_decision_rate()

→ if (ego.capital_ratio() >= Basel.min_car) and (ego.leverage_ratio() >= Basel.min_lev) then

if (re >= 0) and (freeze < ego.freeze_threshold) then

Loan.grant(ego, firm, L0, r_offer, ego.loan_duration)

else // Ration credit to affordable level; reject if none

if (Lmax > 0) then

Loan.grant(ego, firm, Lmax, r_offer, ego.loan_duration)

else

Loan.reject(ego, firm)

end

end

else

Loan.reject(ego, firm)

end where

L0 := firm.requested_amount

r_offer = ego.key_rate() + ego.standing_spread + ego.markup

OCF_e := firm.expected_cashflow() // per-period

PD := 1 / (1 + exp((OCF_e - ego.ub * (r_offer + 1/ego.loan_duration)) / (r_offer + 1/ego.loan_duration)))

LGD := max((L0 - firm.collateral_value()) / L0, 0)

re := Finance.NPV(OCF_e, r_offer, ego.loan_duration) - (PD * LGD * L0)

Lmax = max(0, firm.Affordable_redemption(r_offer)) // credit-ration cap

freeze = Shock Manager.liquidity_freeze() // ~ U[0,1]

The second ML3 rule captures the banks' response when considering firm loan applications under uncertainty and liquidity shocks. The banks extend loans if they are compliant with regulatory standards (capital and leverage ratios) and if the positive expected return on the loan is maintained when default probabilities (PD) and possible losses (LGD) are taken into account.

A liquidity freezes parameter mimics credit tightening in times of financial shocks, with banks rationing or refusing credit despite the solvency of firms. By directly modeling credit rejection and rationing, this rule illustrates how banking sector financial shocks transmit to firms, limiting investment and accentuating systemic uncertainty. This then significantly determines whether firms can maintain value creation or have to contract or bankrupt.

3.5 Investment Market

In addition to household consumption (Section 3.3.3), the second pillar of demand is investment demand, which becomes extremely responsive to macroeconomic uncertainty and financial shocks. The investment market is captured in a non-growing setup in order to consider separately the cyclical effect of shocks, adopting the simplification of Lengnick [55] and Popoyan [30]. Like in the base design, there is **no technological innovation or capital accumulation**, but here investment behavior directly includes uncertainty shocks affecting financing costs, risk perception, and firms' value creation expectations.

directly inconsistent with Section 5.4 when discussed R^D as a key driver

3.5.1 Investment cycle

Within this setup, investment is in accordance with the Keynesian heritage [56], where future demand and possible profits are created through current investment. Uncertainty shocks (i.e., unanticipated liquidity freezes, unexpected interest rate increases) make firms less willing to invest, though. At the micro level, an investment stimulus by one firm triggers demand for others, but during times of increased uncertainty, the multiplier effect fails as firms reduce their reactions.

The rule for the updated amount of targeted investment:

$$\text{if } ret_i > 0 \text{ and } inv_i > 0 \text{ and } inv_i < \lim_i \Rightarrow inv_i^T = inv_i \cdot [1 + \gamma_{inv} \cdot \xi] \quad (10)$$

$$\text{if } ret_i < 0 \Rightarrow inv_i^T = inv_i \cdot [1 - \gamma_{inv} \cdot \xi] \quad (11)$$

$$\text{if } inv_i \geq \lim_i \Rightarrow inv_i^T = \lim_i \quad i \quad (12)$$

Return on investment is given as:

$$ret_i = \text{Turnover}_i - \text{Cost}_i \quad (13)$$

where turnover is sales to other companies and costs are purchases from suppliers. Financial shocks raise the cost of borrowing, lowering ret_i and shrinking liquidity limits, hence investment demand is directly decreased.

Pro-cyclical investment is captured in the transition rate:

$$\text{rate}_i^{inv} = \begin{cases} \text{cap}_i^{-1}, & \text{if } \frac{S_i}{S_i^T} \leq 1 \\ \text{cap}_i^{-1} + S_i \cdot inv_i, & \text{otherwise} \end{cases} \quad (14)$$

where cap_i is the distance from average capacity, S_i/S_i^T is the solvency ratio, and S_i^T/S_i the insolvency barrier (Section 3.3.4). An increase in distance from capacity (slack) decreases investment incidence, whereas a closer distance increases pro-cyclicality. Uncertainty shocks feed in through ξ and through the interest pressure int_i , which is defined as:

$$int_i = \frac{\text{Loan Interest Payments}_i}{\text{Principal Repayments}_i} \quad (15)$$

Increased financial pressure (int_i) or increased uncertainty reduces the investment cycle, expressing value destruction under shocks.

3.5.2 Investment demand

Investments are put in place when companies buy commodities from others. Like in consumption at households, all commodities are homogeneous, though companies' desire to invest is based on liquidity as well as perceived uncertainty. The matching procedure is guided by a stochastic rule like in Section 3.3.3, though purchase incidence is multiplied by $inv_i^{Tinv} \cdot T_{inv}$.

Funding condition: In accordance with the pecking order theory [62], companies' preference goes for internal sources (liquidity, retained earnings) prior to recourse to external debt. In uncertainty shocks, liquidity cushions in precautionary form are the first to alleviate the demand of borrowing. In cases that internal funds are not enough, businesses attempt to seek credit within the market, but external funding is curtailed by shocks such as the freezing of liquidity or more restrictive bank lending requirements. This investment is highly sensitive to uncertainty at the macro level as well as cost stability in the financial sector which subsequently determines whether the companies can continue the generation of economic values or shrink.

3.6 Credit Market

The credit market plays a major role in the transmission between the real economy and the financial sector. In ideal circumstances, banks are the sources of liquidity to aid production and investment by firms and this forms the basis of economic value creation. In macroeconomic uncertainty and financial shocks, though, the same mechanism can support declines through pro-cyclical tightening, rationing and risk aversion. This market organization is therefore necessary to understand the responsiveness of firms and emergence of systemic risks.

3.6.1 Demand for credit

Companies require credit due to two reasons: (i) to finance investment projects and (ii) to offset liquidity shortfalls of wages or working capital. Liquidity-motivated demand is:

$$Liq_i^T = \max(W_i^T - Liq_i, 0) \quad (16)$$

W_i^T , planned wage bill on the prerequisite of production goals, and Liq_i , real liquidity.

Uncertainty shocks influence demand in two respects:

- Precautionary borrowing: companies borrow more than absolutely necessary to maintain liquidity buffers against future shocks.
- Forced refinancing: as revenues fall sharply, companies need more credit to roll over debt, even if profitability is in negative terms.

Both phenomena raise aggregate credit demand in downturns, exactly when banks are more risk-averse. This misalignment generates the credit crunch conditions, amplifying real economy contractions.

3.6.2 Credit supply

Banks decide on loan applications weighing expected return, risk of default, and regulatory constraints. Expected return is:

$$r_{eb} = \sum_{n=1}^{dur} \frac{Pe_i}{(1+iL)^n} - L_{ib}^0 - (pd_{ib} \cdot LGD_{ib} \cdot L_{ib}^0) \quad (17)$$

Default probability rises sharply under shocks, captured as:

$$pd_{ib} = \frac{1}{1 + e^{\frac{OCF_i^e - v_b L L R_i^0}{L H E_i^0}}} \quad (18)$$

where O_i^e = expected operating cash flow and v_b = bank risk attitude. Negative or volatile cash flows raise default probability, driving firms into speculative or Ponzi positions according to Minsky's taxonomy.

Rationing occurs when banks extend only partial credit, compelling companies to reduce investment or output. When there are shocks, however, rationing is not random but discriminatory against poor firms, speeding up bankruptcies and lowering aggregate value creation. Even solvent companies can be rationed if collective bank risk appetite decreases.

Regulatory limits also bind supply. The highest credit capability of banks is:

$$\tilde{C}_b = E_b \cdot \sigma_b \cdot CAR \quad (19)$$

where E_b = equity, σ_b = risk-weighting, and CAR = capital adequacy requirement. During boom periods, improved profits increase E_b , stimulating lending; in busts, losses reduce equity and reduce supply. This is pro-cyclicality that represents the financial accelerator, in which balance sheet weakening leads back to more stringent credit, deepening the downturn.

3.6.3 Firm solvency as determinant of creditworthiness

The creditworthiness of a firm is pegged on a solvency ratio:

$$DR_i = \frac{\sum_{k=1}^n L_k + O_i}{Li q_i + E_i + OCF_i^e} \quad (20)$$

- Low debt ratios portend hedge borrowers, capable of paying up both the principal and the interest.
- Middle-range debt ratios form speculative borrowers, who can only pay interest.
- Ponzi borrowers due to their high debt ratios are firms where new debt is used to pay off old debt.

The risk weighting schemes of banks punish speculative and Ponzi borrowers, however in booms there is over-optimism and effective risk weights are lowered, causing unsustainable credit growth. On the contrary, during periods of slump, pessimism puts risky assets on sterilization by withholding credit even to sound companies, a suicidal pro-cyclicality.

3.6.4 Bank strategy and amplification under uncertainty

Banks change dynamically their interest margins, equity buffers and credit supply. During periods of uncertainty, margins rise, equity is depleted through loan losses and credit potential diminishes. This creates a risk spiral: declining firm profits amplify the number of defaults, which debilitate banks, once again limiting credit. Stabilisation of the system might require government interventions, e.g. bank bail-out or emergency liquidity. Although bailouts keep the banks operating in the lending business; it also brings moral hazard concerns that might be tempting banks to take risks during boom times.

Government - limited bail out to rebuild capital adequacy

Government

| Bank. Any (alter. equity < 0 or alter. equity / alter. RWA < Basel min_ CAR)

@ Policy. Bailout assessment rate ()

-> b = Bank all find (alter. equity < 0 or alter. equity / alter. RWA < Basel min CAR)

Need equity = max (0, Basel min CAR * b. RWA - b equity)

if (need equity > 0) then

 Treasury. Transfer (to = b.id, amount = need equity)

 b. equity += need equity

 Registry. Log ("bailout", bank id = b.id, amount = need equity)

End

The net effect is that the credit market is a value creation mechanism with a supportive role, and a value destruction mechanism in response to shocks. The solvency of firms, rationing of credit and regulatory buffers are not strictly determined by regulators or by the current state of uncertainty as they interact dynamically to create recovery or long-run stagnation.

3.7 Monetary System

The financial system is at the centre of financial shock propagation and the stabilization of value creation in the economy. The essence of this is that although the basic circuit for bank money creation and settlement is the same as under the fractional reserve system, the uncertainty of macroeconomic life fundamentally transforms the operations of reserves circulating, the operations of interbank markets, as well as the work of monetary policy. Within this framework, the monetary system does

not just facilitate settlement of payments, but is also a major amplification mechanism in times of crises as shocks to uncertainty disrupt interbank liquidity and credit supply.

3.7.1 Uncertainty in interbank market

Interbank market facilitates settlement of interbank transactions and supports redistribution of liquidity where they have different volume of transactions. The deposits are endogenously created as banks loan to firms (Section 3.5) and go to households through wages and consumption. Settlement demand reserves which are borrowed and lent to banks at the interbank rate. Uncertainty shocks (e.g., abrupt loss of liquidity, higher counterparty risk) upset this process by causing diminished readiness to lend reserves. Under imbalance of reserves, the interbank interest rate im_{bim_bimb} is dynamic:

$$\text{Text} \quad RE_b = RD_b \text{ and } im_b > i\bar{m} \Rightarrow im'_b = im_b - \gamma_i \cdot \xi \quad (21)$$

$$RE_b = RD_b \text{ and } im_b < i\bar{m} \Rightarrow im'_b = im_b + \gamma_i \cdot \xi \quad (22)$$

$$RE_b < RD_b \Rightarrow im'_b = im_b + \gamma_i \cdot \xi \quad (23)$$

$$RE_b > RD_b \Rightarrow im'_b = im_b - \gamma_i \cdot \xi \quad (24)$$

When banks with insufficient equity to comply with solvency standards are unable to receive interbank loans, the Central Bank steps in as Lender of Last Resort. This provides continuity in payment, but brings to light the vulnerability of the system: during crises, it is common for systems to depend on central bank liquidity, which is expensive and undermines trust. The interbank rate is maintained within a range through open market operations which are set understanding facilities (marginal lending and deposit facility). But when there is a shock of uncertainty, the spreads become wider and the volatility begins to rise and, in this case, the Central Bank has to be more aggressive than before. This echoes the effects of shocks on the reserve distribution mechanism, which increases systemic stress as in the Fig 3.

3.7.2 The mechanism of monetary policy when there is a shock

The monetary policy operates in an inflation-targeting approach which is further extended to accommodate the issue of financial stability. Central Bank is measuring inflation on 12 occasions and comparing it with the target, in addition to tracking the output gap and systemic uncertainty. The key rate is set through an amended Taylor rule:

$$i_c = i^* + \alpha_\pi(\pi - \pi^*) + \alpha_y(y - y^*) + \alpha_u U \quad (25)$$

The extra condition $\alpha_U U$ 2x U 2x, represents tightening or loosening of monetary policy in response to systemic uncertainty: as financing costs and risk spread rise with shocks, the Central Bank reduces rates or provides liquidity, and as systemic uncertainty disappears but inflation explodes, the policy tightens. The potential output y^* is at 95% of full employment, as would be the case under NAIRU (5%). In that context, however, uncertainty shocks may reduce effective potential output by impairing the investment capacity of firms and the willingness of banks to lend, thereby moving the value creation yardstick down.

3.7.3 Stabilization and value generation

The monetary system consequently serves dual purpose (i) to facilitate money flow and payment settlement, and (ii) to affect the firm resilience to value creation in case of a shock. During periods of peace, the interbank lending and activities of the central bank maintain a smooth flow of liquidity. In the face of uncertainty, however, credit frictions, high spreads and precautionary hoarding affect this circulation, worsening recessions. The framework incorporates an explicit uncertainty channel in monetary policy and interbank dynamics, which capture the transmission of shocks between the financial system and firm level behaviour, influencing both survival in the short term and economic value-creation in the long-run.

3.8 Fiscal Mechanism

there is no government budget constraint that matters a lot when you study "fiscal costs in crisis"
→ especially when you claim about "fat-tailed fiscal cost"???

The fiscal mechanism is the input of government actions in the current economic value generation under shocks in stabilizing households, firms and banks. Unemployment is $UB_h = 0.5$ u.w. $L_h U$ is provided to the households to ensure there is minimum consumption demand during the times of labor market weaknesses. **Subsidies targeted** $Sub_i = 0$ $\ln v_i T$ are intended to assist firms in the case when access to credit is limited and allow them to invest where required in the case of financial stress. **Recapitalizations Bailout** $= \max(0, CAR - \min("min" RWA - E - b))$, bail out banks, and save them, preventing collapse of the credit markets. They together produce fiscal multipliers, which make uncertainty shocks less pro-cyclical, with the benefit that they reduce bankruptcy cascades and the long-run capacity to generate economic value, but the disadvantage that they generate potential moral hazards.

4. Calibration

It is essential that the model be calibrated to reflect a realistic representation of the impacts macroeconomic uncertainty and financial shocks have on firm behavior, banking dynamics, and value creation in general. Calibration in this sense has two functions: (i) to assure the stylized business cycle and financial cycle dynamics are reproduced by a set of submodels under stable conditions, and (ii) to add realistic propagation of uncertainty shocks through firms, banks, households, and policy interventions. A mixture of empirical data, stylized facts, and theoretical benchmarks determine parameterization.

4.1 Parameter calibration

The model has several interacting sub modules: the goods market, investment market, credit market, monetary and fiscal mechanism (see Fig. 3). A sensitivity analysis of each submodule is performed based on a one-factor-at-a-time (OFAT) methodology to check theoretical consistency and strength against shocks. Besides this, a plackett-Burman experimental design is used in the goods and investment market sub modules to determine theoretical ranges of behavioral parameters like adjustment speeds in pricing, production, and sensitivity to investment uncertainty. Parameters used in calibration in the credit market to simulate stylized credit cycles, such as the probability of default (Eq. 8), loss-given-default, and credit rationing parameters, are calibrated. The monetary system is also set at the composite model level, in which the modified

Taylor rule (Eq. 13) balances shock mitigation, output stability and inflation control. Specifically, the uncertainty weight α is adjusted to ensure that monetary policy responds sufficiently to systemic stress at the expense of throwing off inflation targeting.

4.2 Initialization

Initialization helps in the specification of initial values of parameters and state variables to reduce warm-up effects and meaningful dynamics of uncertainty. As suggested by Dosi et al. [54], the first values are selected in such a way that they indicate aggregates of economic significance. Mean capacity of production per firm is defined as a ratio of total households to the total firms. Household demand is set at the level of wages and unemployment benefits and firm demand is set at desired levels of investment. Bank equity is tuned to permit credit creation based on Basel capital requirements, but liquidity and inventories are set to close to steady-state values. The uncertainty shocks are modeled by the use of stochastic parameters but with the calibration where the system is in a reasonable equilibrium. Values are chosen sufficiently to reduce the warm-up (100-200 periods) required to stabilize the dynamics before shocking scenarios are studied. The population size of agents is set by scaling ratios based on empirical considerations to keep the number of firms to banks and households realistic: four agents per bank and fifty households per firm (e.g., 12 banks, 48 firms, and 480 households). Central bank and a government are among the distinctive agents. Sensitivity analysis establishes that the behaviour of the basic model is not affected by different population sizes between 10 and 200 firms, but very small populations does enhance stochastic noise. Since simulator is limited, we fail to test population greater than 200 firms. To be validated, the base model of 12 banks, 48 firms, 480 households, one central bank, and one government is simulated over 600 periods, which represents several business and financial cycles with periods of shocks and recovery in alternating phases.

5. Validation

The objective of the model is to understand how macroeconomic uncertainty and financial shocks may impact firm behavior, credit behavior, and the creation of economic value, as well as to examine the efficacy of institutional and policy responses. One way to determine if the model output is reliable to generate policy experiments is to determine whether the model can simultaneously generate the broad set of macroeconomic and microeconomic stylized facts [69]. A strictly history-friendly model [22], in which the model is closely adjusted to historical data, is inappropriate here because the counterfactual nature of simulated uncertainty shocks and financial crises. Rather, validation is based on correspondence of stylized facts describing business cycles, financial cycles and firm-level dynamics during shocks. It is verified against a chosen list of macroeconomic and financial stylized facts (SF), which is extended with markers of uncertainty shocks and value generation ability (Table 1). These are volatility dynamics of GDP, consumption, and investment, the association between firm indebtedness, credit rationing, and defaults, counter-cyclicalities of bankruptcies, skewed distributions of banking crisis duration and fiscal costs, and firm-level investment lumpiness, and persistent unemployment. Besides this, we have system-wide policies like the Phillips curve, the Okun law and interbank market forces. The emphasis is on cyclical changes and resilience processes in a context of uncertainty as opposed to long-run growth facts as the latter would be nonexistent because of the no-productivity-growth assumption.

Table 1. Stylized facts reproduced in the model

Code		Selected empirical references
SF1	Relative volatility across output, consumption, and investment	(Stock & Watson, 1999)
SF2	Duration of recessions follows an exponential distribution	(Wright, 2005)
SF3	Cross-correlations among key macro variables (GDP, credit, investment)	(Mauro Napoletano, 2017)
SF4	Investment at the firm level moves pro-cyclically, especially under shocks	(Wälde & Woitek, 2004)
SF5	Credit cycles: indebtedness and defaults show lagged correlation	(Lown & Morgan, 2006)
SF6	Strong co-movements between credit spreads and uncertainty indices	(Mendoza & Terrones, 2012)
SF7	Banking crisis episodes show right-skewed duration	(Reinhart & Rogoff, 2009)
SF8	Fiscal costs of crises (to GDP) are distributed with fat tails	(Laeven, n.d.)
SF9	Firm-level investment is lumpy and discontinuous	(Doms & Dunne, 1998)
SF10	Corporate bankruptcies are counter-cyclical and intensified by shocks	(Jaimovich & Floetotto, 2009)
SF11	Phillips's curve: negative relation between unemployment and inflation	(Lipsey, 1960)
SF12	Okun's law: output gap is linked to unemployment fluctuations	(Lodewijks, 2016)

SF13	Interbank market shows stress propagation and liquidity freezes	(Bech & Monnet, 2016)
SF14	Unemployment tends to persist after financial shocks	(Lavoie, 2015)
SF15	Government interventions (subsidies, transfers, bailouts) stabilize cycles but may create moral hazard	(Ricchetti et al., 2015)

Micro-founded firm-level behaviors are aggregated to support aggregate results. One of the key macroeconomic indicators of output is Gross Domestic Product (GDP) which measures the **total value of production of goods and services** including household consumption, corporate investment, and inventory under shock conditions. Here, the (real) GDP is defined as:

$$GDP^{\text{real}} = \sum_{i=1}^n q(C_i + I_i + \Delta N_i) \quad \text{this is not value creation in finance or economics} \quad (26)$$

C_i household consumption, I_i firm investment, and ΔN_i change in inventories of firm i . Analogous to the calculation of nominal GDP, firm-specific prices are calculated. The uncertainty shocks enable GDP fluctuations reflect both endogenous cycle and exogenous stress propagation. In extracting the business cycle component of simulated time series, the Baxter-Kling filter is used with default parameters: $\text{bpf}(6,32,12)$, the cycles 632 quarters and the lag/lead length of 12. The model periods are one quarter in number, that is, 400 model periods would represent 100 years. This configuration allows cyclical properties to be sustained through repeated shocks. Each configuration has 20 replications, which validate the results. Mean SD of real GDP among replications is extremely small ($\rho = 3.471\text{e-}04$) and implies consistent replication of cyclical processes. Replication also guarantees that results like the transmission of financial distress, credit rationing, and fiscal stability are not single run artifacts. It can be extended in the future with the use of transient analysis and confidence-level estimation to fine-tune the number of replications required to interpret particular policy situations.

5.1 Relative volatility of GDP, investment, and consumption — SF1

The first stylized fact measures the relative volatility of both GDP, investment and consumption in the impact of macroeconomic uncertainty shocks. Volatility in this context is both a result of dynamic business cycle processes and is magnified by demand shocks, liquidity freezes and changes in policy credibility. As shown in figure 6, the model output captures the empirical regularity that investment is more volatile than GDP, and consumption is relatively stable. The relative standard deviation of consumption to GDP is 0.47, investment 4.75 and unemployment 14.10 with 20 repetitions. Such values are in line with empirical standards [70, 71, 86], in which investment is generally three to five times more volatile than GDP and unemployment is strongly cyclically amplified as indicated in the Fig 4. The consumption on the other hand has less variability as it carries stabilizing fiscal transfers and precautionary savings in response to uncertainty shocks.

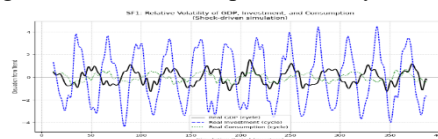


Fig. 4. Relative Volatility of GDP, Investment, and Consumption (Shock-driven Simulation)

This model also reveals that larger uncertainty shocks congest consumption by inducing precautionary consumption but increases investment cycles volatility. Calibration experiments that involve more fiscal stabilization (e.g., higher levels of unemployment benefits) restrain unemployment volatility but have less impact on investment changes. As with earlier results [70, 87, 88], the model exhibits long-term cyclical effects without overall growth patterns, indicating the shock-driven, as opposed to productivity-driven, character of the simulated economy.

5.2 Time of recession exponentially distributed - SF2

In this context, recession is taken to mean a series of quarters where the growth of real GDP is negative. In the presence of uncertainty shocks, e.g., abrupt demand contractions, or freezes in liquidity, or uncertainty about policies, the frequency of recessions and their length distribution are stylized, with the cumulative distribution of length being well decayed towards the left, i.e., exponentially distributed, as observed in empirical data. The simulated economy replicates this stylized fact, with the cumulative duration distribution strongly exponentially distributed. In 20 replications, the majority of recessions span 1-6 periods, with some longer crises spanning 10-13 periods, with some of these being related to major financial shocks. This is consistent with empirical evidence that most recessions are temporary, whereas systemic financial crises greatly enhance the tail risk of long recessions. Therefore, this model explains the exponential distribution property with the amplification of recession persistence by macroeconomic uncertainty.

5.3 Macro variable cross-correlations- SF3

The third stylized fact is about the structure of cross-correlation of macroeconomic variables when subjected to financial shocks and financial uncertainty. In Table 2, investment and inflation are highly pro-cyclical, whereas unemployment is counter-cyclical as Waalde and Woitek explain. Notably, investment is the driver of GDP since future decisions on investment adjustments are made by firms in line with uncertainty expectations and credit conditions, and output is the lagged adjustments such as labor and production capacity as indicated in the Fig 5.

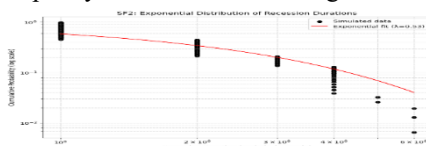


Fig. 5. Exponential Distribution of Recession Durations

Inventories respond to GDP changes, indicating that firms hoard precautionarily in the face of uncertainty shocks before production is altered. During boom periods, the effect of inflation is positively correlated with output because it is measured

across several periods but lags in response. Unemployment demonstrates a significant negative correlation with GDP, which represents the magnified labor market impacts of financial stress. In total, the model captures major empirical co-movements: investment cycles are followed by output changes, inventories are slow to change, pressure on the demand side is followed by inflation, and unemployment is counter-cyclical. These patterns indicate that the model recreates the transmission of uncertainty shock by the real and financial sectors illustrated in the Fig 2.

Table 2. Cross-correlation of macro variables — SF3

Series (Bpf)	t-4	t-3	t-2	t-1	t	t+1	t+2	t+3	t+4
GDP	0.20	0.46	0.72	0.92	1.00	0.92	0.73	0.52	0.19
Investment	0.51	0.32	0.07	-0.19	-0.44	-0.63	-0.78	-0.82	-0.74
Change in Invent.	-0.27	0.00	0.30	0.58	0.93	0.78	0.52	0.24	-0.05
Inflation	0.00	0.19	0.36	0.58	0.66	0.71	0.66	0.47	0.18
Prices	0.42	0.56	0.66	0.71	0.90	0.89	0.68	0.57	0.40
Unemployment	-0.35	-0.55	-0.70	-0.82	-0.82	-0.75	-0.50	-0.24	0.07
GDP - Δ Invent.	0.36	0.66	0.90	0.98	1.00	0.87	0.69	0.37	0.08

5.4 Pro-cyclical aggregate R&D investment — SF4

Here, the aggregate firm R&D investment has a sharply pro-cyclical pattern which supports the connection among uncertainty, innovation and long-term value creation. According to figure 8, in case of an increase in GDP, the increase in firms increases their R&D expenditures as a growth strategy and to take advantage of favorable financing environment. On the other hand, in the case of uncertainty shock, like liquidity freeze or policy turmoil, companies reduce spending on R and D, which further worsens depressions as illustrated in Fig 6.

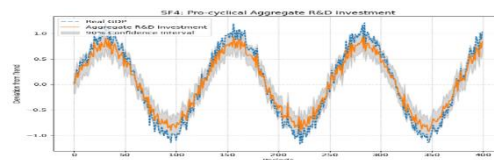


Fig. 6. Pro-cyclical aggregate R&D investment

Notably, this model has shown that the fluctuations in the GDP are caused by the fluctuations in the R&D investment, which shows the forward-looking behaviour of the firms when there is uncertainty. During boom times, more credit is available and innovation-based projects are supported, whereas during recessions, companies transition into a liquidity-saving mode, limiting long-term growth opportunities.

5.5 Cross-correlation of credit variables — SF5

The cyclical nature of credit related variables is exaggerated by uncertainty shocks. Bank profits and firm debt are pro-cyclical as would be expected. Table 3 illustrates that bank profits follow GDP by one period because of delay in loan repayment timelines, whereas firm debt follows closely output as investment and liquidity funding increases with the economy.

Table 3. (Cross-correlation of Credit Variables — SF5)

Series (Bpf)	t-4	t-3	t-2	t-1	t	t+1	t+2	t+3	t+4
GDP (real)	0.20	0.46	0.73	0.93	1.00	0.92	0.73	0.46	0.19
Bank profits	-0.04	0.23	0.49	0.72	0.72	0.54	0.31	0.54	0.53
Firm debt	0.09	0.23	0.47	0.71	0.91	0.86	0.73	0.31	0.21
Credit defaults	0.08	0.07	0.03	-0.19	-0.44	-0.63	-0.78	-0.82	-0.74

Credit defaults, on the other hand, are counter-cyclical and peak during downturns when the anticipated profits and cash flows of firms plunge. In extreme financial shocks, the number of defaults surges and the equity buffers of banks are diminished, making the credit tightening even more extreme. This tightening that is pro-cyclical increases macroeconomic volatility because the ability of firms to take loans decreases at the time when it is least required.

5.6 Cross-correlation between loan loss and firm debt- SF6

The interplay between the accumulation of firm debt, and loan losses is based on a leverage-driven dynamic. Figure 9 shows that firm debt increases in boom times but ultimately causes an increase in loan losses in the next downturns, suggesting that the lagged positive correlation is supported as in Fig 7. The tendency is consistent with a Minskian leverage cycle, in which because of high debt ratios, they are vulnerable to uncertainty shocks.

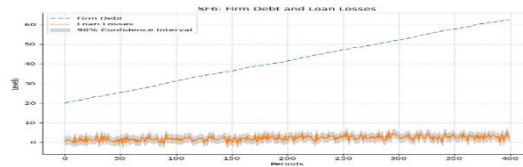


Fig. 7. Firm debt and loan losses

During uncertainty shocks, highly leveraged companies encounter solvency strains, and this results in defaults and write-downs on bank balance sheets. This destroys the capacity to supply credit, extends recessions and diminishes the creation of economic values. The model therefore underlines the interplay of financial weakness and macroeconomic uncertainty in contributing to systemic risk.

5.7 The duration of bank crisis is skewed to the right SF7

A banking crisis is considered to be the successive phases of one or more bank failures. The model yields a skewed right distribution of crisis periods under uncertainty shocks (e.g., liquidity freezes, policy uncertainty) as found in the Fig 8 in the Reinhart and Rogoff. The majority of banking crises are brief, whereas some are more prolonged in instances of co-existing shocks in combination with high leverage and narrow equity cushions, which is a tail risk that damages value creation by creating sustained credit dysfunction.

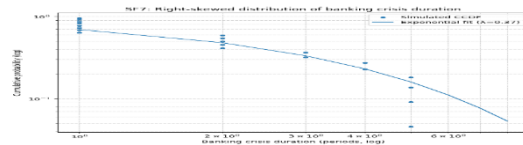


Fig 8. (SF7)- Banking crisis duration distribution

5.8 Fiscal cost of banking crises is fat-tailed — SF8

Simulated fiscal cost-to-GDP ratio is fat-tailed and excess kurtosis in accordance with Laeven and Valencia. Normality is rejected by JarqueWilk and ShapiroWilk tests. Tail realizations occur in the model when crises coincide with tight interbank liquidity and a high rate of firm default, exactly where the availability of public backstop borderline collapse at high fiscal cost.

5.9 Lumpy investment on the firm level — SF9

Firm investment is lumpy: periods of stagnation interspersed with bursts, particularly in the case of sensitive to uncertainty and credit-to-outside sources R and D, the financing of which is lumpy. The three prototypical firms exhibit clustered spikes spaced by low (or no) investment spurts, indicating precautionary liquidity hoarding in periods of stress and opportunistic scaling in recoveries-both important processes underlying heterogeneous value creation as in the Fig 9.

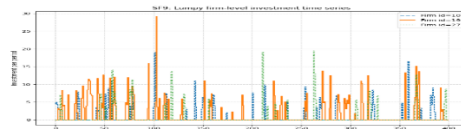


Fig. 9. (SF9) Lumpy firm-level investment (three firms)

5.10 Firm bankruptcies are counter-cyclical — SF10

Bankruptcies increase during downturns and dominate the real GDP by approximately two periods in our simulations. Uncertainty shocks squeeze cash flows and narrowed credit, forcing speculative/Ponzi firms (high debt ratios) into bankruptcy. This contracyclic pattern of bankruptcy is an indicator of turning points (boom→bust), diminished productive capacity, and losses are transmitted to banks, and this lowers value-generation as depicted in the Fig 10.

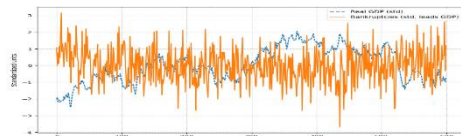


Fig. 10. (SF10) Bankruptcies are counter-cyclical (lead/lag + overlay)

5.11 Phillips curve — SF11

The model is similar to a downward-sloping Phillips curve, with more unemployment corresponding to less inflation (inflation is calculated with three periods). When there is uncertainty spike, demand compression causes the short-run trade-off to shift and flatten the slope; policies of stabilization then assist in re-anchoring the inflation-unemployment relationship as in Fig 11.

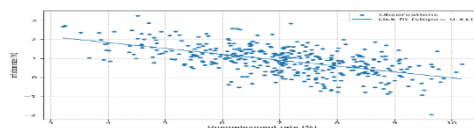


Fig. 11. (SF11) Phillips's curve (inflation vs unemployment)

5.12 Okun's law — SF12

the difference form of Okun law: the relation of change of unemployment to real GDP is negative. In the presence of uncertainty, the slope becomes steeper as the firms respond more quickly to the uncertainty through precautionary layoff; the effects of fiscal transfers and credit support producers maintain the value creation capacity as shown in the Fig 12.

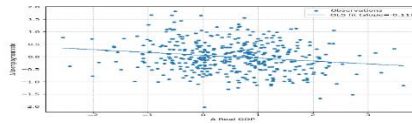


Fig. 12. (SF12) Okun's law (difference version)

5.13 Interbank market — SF13

Three stylized facts are obtained on a liquidity-injection experiment (Central Bank adds fixed reserves per bank in period 50-100): (i) the interbank rate decreases toward the lower bound of the corridor; (ii) the interbank volume decreases with increasing reserves; (iii) the volatility of the rate decreases. These impacts are larger and more lasting under uncertainty shocks and they emphasize the policy liquidity role in alleviating funding strains and facilitating credit provision to companies as it can be seen in Fig 13.

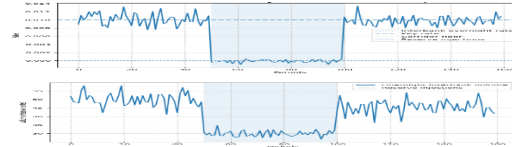


Fig. 13. (SF13) Interbank experiment (rate → floor; volume ↓; volatility ↓)

5.14 Discussion

As shown in the validation results, the proposed agent-based model is capable of replicating a wide scope of macroeconomic and financial stylized facts in the presence of uncertainty shocks. This model is not only the pro-cyclicality of investment, credit and bank profits, but also the counter-cyclicality of unemployment and bank failures. Such findings are consistent with empirical studies and prove the usefulness of the model as a tool to test how firm behavior, financial health, and macroeconomic stability are related. Importantly, the amplifying effect of uncertainty shocks is evident: volatility rises, recession durations become extended and the likelihood of systemic crisis rise when the shocks are complemented by excessive leverage and low liquidity levels. Among the contributions of the model is the fact that it bridges the gap between the heterogeneity of firms and the dynamics at the macro-level. The micro-level abnormalities that occur because of lumpiness of firm investments (choices about R&D and financing accessibility) add up to cyclical variation. Similarly, the contagion of defaults within the banking system is also the reason why the solvency shock of individual actors can produce systemic crises. This model alone generates doubtfulness in the body of demand and financial structure and the point is that an economy is endogenously vulnerable to abrupt shocks in such a way that firm-level decision rules are coupled with macro-financial resilience. As per the analysis, it is also clear how institutional and policy interventions come in handy in curbing systemic instability. The fiscal transfer and unemployment benefits normalize the household consumption and central bank liquidity injections, without the interbank market freeze in the long-term. However, this model has trade-offs: stabilization will reduce volatility in the short-term, but will also increase moral hazard since firms and banks will anticipate governmental and central bank bailouts. This conflict demonstrates that there is necessity to come up with policies that would help to minimize the disaster in the short term and to sustain the motivation to act long-term in the sustainable fashion that will generate value.

6. Conclusion and Future Work

Using the agent-based simulation framework and macroeconomic uncertainty, this research demonstrates that the main stylized business and financial cycle facts can be accurately modeled. The framework sheds light on how uncertainty responds to investment patterns, increases credit constraints, and creates systematic weakness as a firm reacts explicitly to financial shocks. The results prove that the model is valid to analyze the interaction between uncertainty and the dynamics of firms and the macro-financial outcomes. There are a few ways that the model needs to be expanded in future. First, the dynamics of productivity and advancement in technology can be combined in order to examine long-run growth consequences of uncertainty and financial shocks. This would enable the study of the effects of innovation and R&D and firm entry-exit dynamics on resilience and value creation beyond short-term cycles. Second, the distribution of incomes, precautionary savings behavior and heterogeneous consumption propensities would give a better picture of the amplification of the shock on the demand side. Lastly, macroprudential instruments and adaptive policy rules can be included in the policy space. To illustrate, some of the policies that have the potential to be tested in the model to determine whether they can work to reduce systemic risk and create sustainable value would include countercyclical capital buffers, uncertainty-adjusted interest rate rules and targeted fiscal support of innovation. These additional extensions would enhance the framework to be a stress-testing policy strategies laboratory in the presence of uncertainty-related crisis, a gap between microeconomic behaviour and macroeconomic performance.

Declaration

Data Availability

The data used in this study are generated from the agent-based simulation model developed by the author. All simulation codes and datasets are available from the corresponding author upon reasonable request.

Conflicts of Interest

The author declares that there are **no conflicts of interest** regarding the publication of this manuscript.

Funding Statement

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Author Contribution

Tianyu Tu is solely responsible for the conception of the study, development of the agent-based model, simulation execution, data analysis, interpretation of results, and manuscript writing.

Ethical Approval

This article does not involve human participants, animals, or sensitive personal data. Therefore, ethical approval was **not required**.

Consent to Participate

Not applicable. No human participants were involved in this study.

Consent to Publication

The author consents to the publication of this manuscript.

Competing Interests

The author declares that there are **no competing interests** that could influence the work reported in this paper.

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References

- Akhatova, A., Kranzl, L., Schipfer, F., & Heendeniya, C. B. (2022). Agent-based modelling of urban district energy system decarbonisation—A systematic literature review. *Energies*, 15(2), 554.
- Bech, M., & Monnet, C. (2016). A search-based model of the interbank money market and monetary policy implementation. *Journal of Economic Theory*, 164, 32–67. <https://doi.org/10.1016/j.jet.2015.08.007>
- Demopoulos, N. (2025). Randomness and structure in socio-economic systems. Preprints. Retrieved from https://www.preprints.org/manuscript/202505.2286/download/final_file
- Doms, M., & Dunne, T. (1998). Capital adjustment patterns in manufacturing plants. *Review of Economic Dynamics*, 1(2), 409–429. <https://doi.org/10.1006/redy.1998.0011>
- Dorrah, D. H., & McCabe, B. (2024). Integrated agent-based simulation and game theory decision support framework for cash flow and payment management in construction projects. *Sustainability*, 16(1), 244.
- Gambino, M., Cuturi, C., Guadalupi, L., & Capasso, S. (2024). Socio-economic analytical frameworks for marine spatial planning: Evaluating tools and methodologies for sustainable decision making. *Sustainability*, 16(23), 10447.
- Giannakopoulos, N. T., Terzi, M. C., Sakas, D. P., Kanellos, N., Toudas, K. S., & Migkos, S. P. (2024). Agro-economic indexes and big data: Digital marketing analytics implications for enhanced decision making with artificial intelligence-based modeling. *Information*, 15(2), 67.
- Güven, D., Kayalica, M. O., & Sen, O. L. (2025). The impact of electricity generation on CO₂ emissions in Türkiye: An agent-based simulation approach. *Energies*, 18(3), 655.
- Haynes, P., & Alemna, D. (2022). A systematic literature review of the impact of complexity theory on applied economics. *Economies*, 10(8), 192.
- Ionescu, Ș., Chiriță, N., Nica, I., & Delcea, C. (2023). An analysis of residual financial contagion in Romania's banking market for mortgage loans. *Sustainability*, 15(15), 12037.
- Ionescu, Ș., Delcea, C., Nica, I., Dumitrescu, G., Simion, C.-E., & Cotfas, L.-A. (2025). Exploring complexity: A bibliometric analysis of agent-based modeling in finance and banking. *International Journal of Financial Studies*, 13(2), 65.
- Jaimovich, N., & Floetotto, M. (2009). Firm dynamics, markup variations, and the business cycle. Social Science Research Network. <https://papers.ssrn.com/abstract=1420966>
- Keenan, J. M., Trump, B. D., Hynes, W., & Linkov, I. (2021). Exploring the convergence of resilience processes and sustainable outcomes in post-COVID, post-Glasgow economies. *Sustainability*, 13(23), 13415.
- Laeven, L., & Valencia, F. (2025). Systemic banking crises: A new database. IMF Working Paper. Retrieved from <https://www.imf.org/en/Publications/WP/Issues/2016/12/31/Systemic-Banking-Crises-A-New-Database-22345>
- Lavoie, M. (2015). Book review: Carlin, Wendy and David Soskice (2015): *Macroeconomics: Institutions, Instability, and the Financial System*. *European Journal of Economics and Economic Policies*. <https://doi.org/10.4337/ejeep.2015.01.10>
- Lipsey, R. G. (1960). The relation between unemployment and the rate of change of money wage rates in the United Kingdom, 1862–1957: A further analysis. *Economica*, 27(105), 1. <https://doi.org/10.2307/2551424>
- Lodewijks, J. (2016). Arthur M. Okun: Economics for Policymaking. *Journal of Economic Surveys*. <https://doi.org/10.1111/j.1467-6419.1988.tb00046.x>
- Lown, C. S., & Morgan, D. P. (2006). The credit cycle and the business cycle: new findings using the Loan Officer Opinion Survey. *Journal of Money, Credit, and Banking*, 38(6), 1575–1597. <https://doi.org/10.1353/mcb.2006.0086>
- Ma, C., Zhang, L., You, L., & Tian, W. (2024). A review of supply chain resilience: A network modeling perspective. *Applied Sciences*, 15(1), 265.
- Martins, A. C., Pereira, M. de C., & Pasqualino, R. (2023). Renewable electricity transition: Evaluating infrastructure investments through real options analysis in Brazil. *Sustainability*, 15(13), 10495.
- Napoleitano, M. (2017). Are business cycles all alike? A bandpass filter analysis of Italian and U.S. cycles. ResearchGate. Retrieved from <https://www.researchgate.net/publication/24132336>
- Mazzetto, S. (2024). Interdisciplinary perspectives on agent-based modeling in the architecture, engineering, and construction industry: A comprehensive review. *Buildings*, 14(11), 3480.
- McGarraghy, S., Olafsdottir, G., Kazakov, R., Huber, É., Loveluck, W., Gudbrandsdottir, I. Y., Čechura, L., Esposito, G., Samoggia, A., & Aubert, P.-M. (2022). Conceptual system dynamics and agent-based modelling simulation of interorganisational fairness in food value chains. *Agriculture*, 12(2), 280.
- Melina, G., Sukono, Napitupulu, H., & Mohamed, N. (2023). A conceptual model of investment-risk prediction in the stock market using extreme value theory with machine learning. *Risks*, 11(3), 60.
- Mendoza, E. G., & Terrones, M. E. (2012). An anatomy of credit booms and their demise. Social Science Research Network. <https://papers.ssrn.com/abstract=2147090>
- Oztemel, E., & Isik, M. (2025). A systematic review of intelligent systems and analytic applications in credit card fraud detection. *Applied Sciences*, 15(3), 1356.
- Peykani, P., Sargolzaei, M., Tanasescu, C., Shojaie, S. E., & Kamyabfar, H. (2025). Investigating the relationship between liquidity risk, credit risk, and solvency risk in banks. *Economies*, 13(5), 139.
- Reinhart, C. M., & Rogoff, K. S. (2009). The aftermath of financial crises. *American Economic Review*, 99(2), 466–472. <https://doi.org/10.1257/aer.99.2.466>
- Rezaei, A., Abdellatif, I., & Umar, A. (2025). AI and ML techniques improving stock market movement accuracy. *International Journal of Financial Studies*, 13(1), 28.
- Riccetti, L., Russo, A., & Gallegati, M. (2015). An agent-based decentralized matching macroeconomic model. *Journal of Economic Interaction and Coordination*, 10(2), 305–332.
- Serkov, L., & Krasnykh, S. (2024). Peculiarities of economic agent behavior under cognitive constraints in a semi-open New Keynesian model. *Mathematics*, 12(1), 95. <https://doi.org/10.3390/math12010095>
- Simmons, R., Dini, P., Culkun, N., & Littera, G. (2021). Crisis and the role of money in real and financial economies. *Journal of Risk and Financial Management*, 14(3), 129. <https://doi.org/10.3390/jrfm14030129>

- Stock, J. H., & Watson, M. W. (1999). Business cycle fluctuations in U.S. macroeconomic time series. *Handbook of Macroeconomics*, 1, 3–64. [https://doi.org/10.1016/S1574-0048\(99\)01004-6](https://doi.org/10.1016/S1574-0048(99)01004-6)
- Sun, W., An, Y., & Gao, Y. (2025). Systemic risk contagion in China's financial–real estate network. *Fractal and Fractional*, 9(9), 557.
- Wälde, K., & Woitek, U. (2004). R&D expenditure in G7 countries: Implications for endogenous fluctuations and growth. *Economics Letters*, 82(1), 91–97. <https://doi.org/10.1016/j.econlet.2003.07.014>
- Wang, D., & He, Y. (2024). Mathematical simulation of South Korea's financial and economic impacts from real estate bubbles. *Mathematics*, 12(19), 3058.
- Wright, I. (2005). The duration of recessions follows an exponential law. *Physica A: Statistical Mechanics and Its Applications*, 345(3–4), 608–610. <https://doi.org/10.1016/j.physa.2004.07.035>
- Zhou, Z., & Shen, Y. (2024). Free riding of vehicle companies under dual-credit policy: An agent-based system dynamics model. *World Electric Vehicle Journal*, 15(6), 227.